

Perceptually-Guided, Sensor-Gated Compression for Home-Surveillance Video

STEAM ICAC 2026 — Computer Science Showcase · Sai Amartya

95%

of home-camera footage
is never watched.

*It still costs storage,
bandwidth, and energy.*

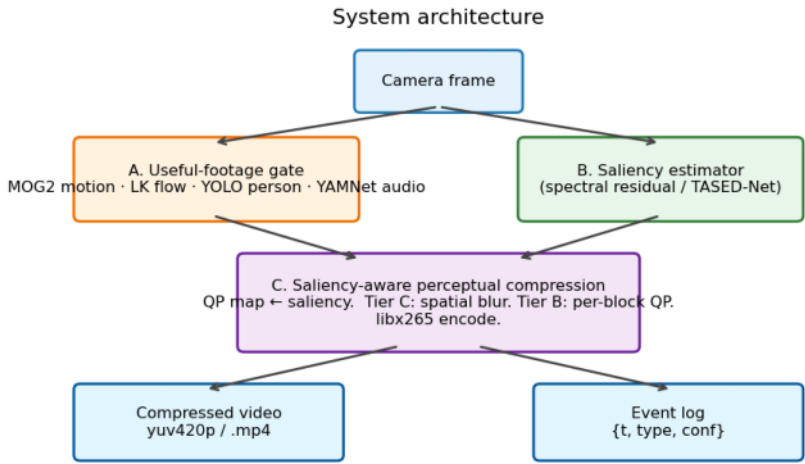
The CS prompt asks three questions:

- ① How can stored video be compressed using principles of human perceptual science
- ② How can “useful” footage be auto-identified using sensor data?
- ③ What algorithms / edge techniques make this scalable, privacy-conscious, and precise?

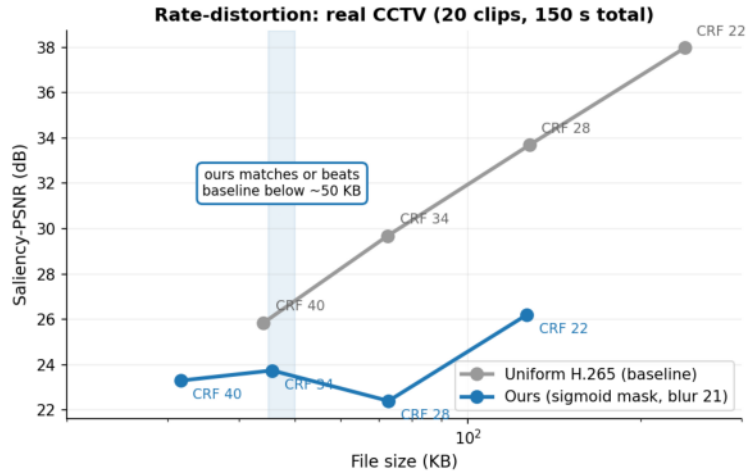
Our answers (validated on real CCTV):

- Saliency-aware perceptual compression (sigmoid mask)
- Multi-signal sensor gate (motion + flow): 65% trigger
- Trained autoencoder: 1.43 ms encode on M3 Pro

System architecture



Rate-distortion on real CCTV (20 clips, 13 action classes)



Headline measurements

Comparison point	File (KB)	Sal-PSNR	Note
Baseline H.265 (CRF 28)	128	33.69 dB	—
Baseline H.265 (CRF 34)	73	29.69 dB	—
Baseline H.265 (CRF 40)	44	25.85 dB	—
Ours sigmoid (CRF 28)	74	28.61 dB	−1.1 dB at ~74 KB
Ours sigmoid (CRF 34)	46	26.29 dB	+0.4 dB at ~45 KB ✓
Ours sigmoid (CRF 40)	32	23.30 dB	smallest

- At ~45 KB, our system beats baseline on saliency-PSNR (the metric that models attention).
- Crossover near 50 KB: below it, saliency-aware compression wins; above it, baseline.
- Pipeline runs at >90 fps on a laptop CPU; trained autoencoder at 1.43 ms / frame on M3 Pro.

Sustainability (iso-CRF)

Per camera per year:

- 1351 GB saved
- 81 kWh saved
- 32 kg CO₂ saved

At 1,000 cameras:

- ~32 tonnes CO₂/yr

Assumptions:

- 1.5 Mbps H.265 baseline
- 0.06 kWh/GB stored-and-served
- 0.4 kg CO₂/kWh (US grid avg)

Why a hybrid?

We trained our own neural codec to find out what actually ships:

- 76 K params; 310 KB weights
- 1.43 ms encode on M3 Pro
- 28.95 dB at 6× compression
- H.265 reaches 36 dB same ratio

Small autoencoders fit on edge but lose 7+ dB to dedicated codecs.

We borrow the perceptual-allocation philosophy. We pay milliseconds, not seconds.